

Q1: a) Solve the time and space complexity of the following loops?

i) for (int i = 1; i <= n; i++) {  
     int j = i;  
     while(j > 0) {  
         j--;  
     }  
 }

Example 5  
 when i = 1  
 1 0 2

(3+3+3 marks)  
 { 1, 2, 3, 4, 5, 6  
 } 1 2 3 4 5 6

ii) while(i > 0) {  
     for(int j = 0; j < i; j++) {  
         statement;  
     }  
     i /= 2;  
 }

iii) for(int i = n; i > 0; i--) {  
     int j = n;  
     while(j > i)  
     {  
         j--;  
     }  
 }

b) Use the master theorem to solve the following recurrences.

(3+3 marks)

- i)  $T(n) = T(n-1) + \sqrt{n}$   
 ii)  $T(n) = 2T(n/2) + n \log n$

c) Use the recursion tree method and find time complexity of the recurrence relation. (5+5 marks)

i) Algorithm recur(int n):

```
for(int i = 1; i <= n; i++) {
    for(int j = 1; j <= i; j++) {
        statement
    }
}
```

ii)  $T(n) = T(n/2) + T(n/4) + n$

$$T\left(\frac{n}{2}\right) = T\left(\frac{n}{2^2}\right) + T$$

Q2: a) Illustrate the following algorithms and its recurrences. Explain which part of the algorithm is represented by each value mentioned in the recurrence relation. (4\*3 marks)

|                                  |                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------------------------------|
| Merge Sort                       | $T(n) = T(k) + T(n-k) + O(n)$<br><i>Where k: <math>T(n)</math> cons has <math>T(n-k)</math></i> |
| Strassen's Matrix Multiplication | $T(n) = T(k) + T(n-k) + O(n^2)$                                                                 |
| Fibonacci                        | $T(n) = T(n-2) + T(n-1) + O(1)$                                                                 |
| Heapify (Build Heap)             | $T(n) = T(n/2) + O(1)$                                                                          |

b) Analyze different asymptotic notations used in algorithm analysis for any one algorithm you know and explain the significance of asymptotic notations. (2+6 marks)

c) Describe the PARTITION() algorithm of Quick Sort and Analyze step by step how you would get the pass1 result by taking last element as pivot on the following array.

|   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|
| 8 | 2 | 1 | 5 | 6 | 1 | 3 | 7 | 4 | 9 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|

Infer the average case time complexity of quick sort.

(15 marks)

Q.3: a) A graphics engine needs to perform matrix transformations on an image represented as a 4x4 pixel transformation matrix. (10 marks)

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{bmatrix}$$

Explain Strassen's Algorithm to multiply these matrices step by step and describe why Strassen's algorithm is better for large image processing tasks compared to the standard matrix multiplication.

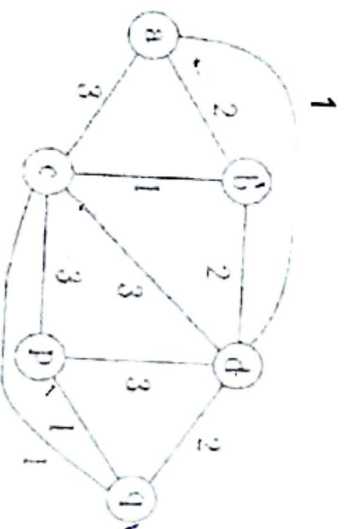
b) Construct an algorithm to solve a CPU scheduling that needs to execute tasks based on given priority (higher priority first).

|    |    |
|----|----|
| T1 | 4  |
| T2 | 10 |
| T3 | 3  |
| T4 | 5  |
| T5 | 1  |

(10 marks)

Q.4: a) Construct the adjacency list of the following weighted graph using 'node a' as source node, and demonstrate on it the Kruskal's algorithm for minimum spanning tree. In the final step, you should write the MST and its cost. Is this MST unique for this graph?

(10 marks)



b) Use prim's algorithm to find out the minimum cost spanning tree and contrast both trees. (10 marks)